

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2036

J R Whiting 100 kV
February 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2036. The Interconnection Customer proposes a 114.3 MW solar photovoltaic facility interconnecting at the J R Whiting 100 kV substation in the Michigan area. The request was received on March 18, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 711 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$998,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Lakeport / Fermi 500 kV, the Lewisville Switch / Doyle Energy Facility 100 kV loads to 105.2% of its 220 MVA rating. Project DFAX: 6.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lewisville Switch / Doyle Energy Facility 100 kV	Reconductor line	\$998,000	100% (\$998,000)
	Total allocated to project		\$998,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.